



ITM SKILLS
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Railway Reservation System

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Problem Statement

Railway Reservation System

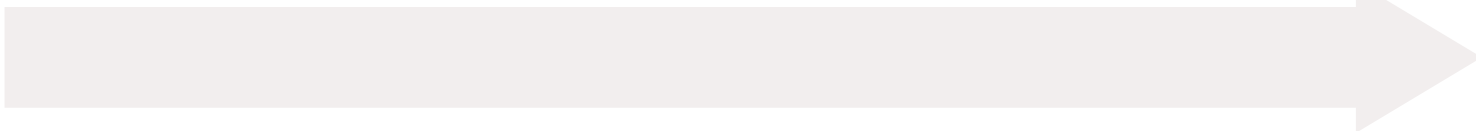
The objective of this project is to develop a simple Railway Reservation System using C++ that addresses the core functionalities of a booking platform.

- Displays available trains with essential details.
- Allows users to effortlessly book seats based on availability.
- Enables hassle-free cancellation of previously booked seats.
- Maintains real-time accuracy of seat availability.
- Generates a comprehensive billing section for all booked seats.

The system operates as a **menu-driven console application**, providing an intuitive interface and running continuously until the user explicitly chooses to exit.

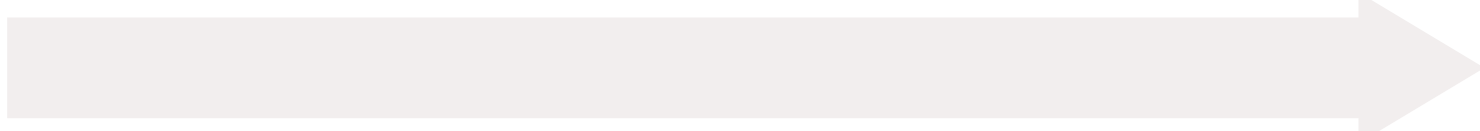


What SOLUTION ? : Railway Reservation Flow



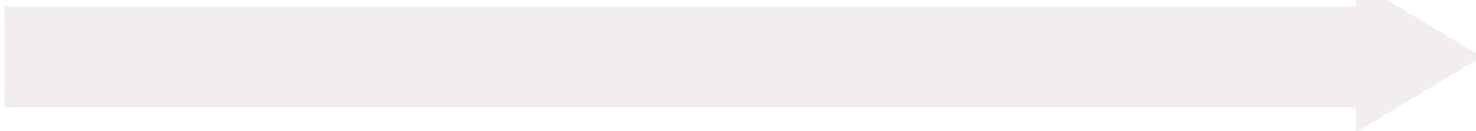
User Selects Option

The user interacts with the menu, choosing actions like booking a new ticket, cancelling an existing reservation, or inquiring about train schedules.



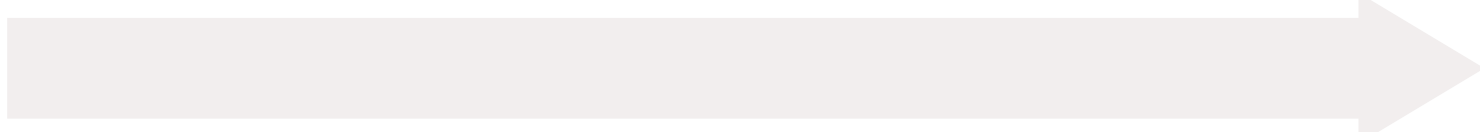
System Validates Request

The system checks for seat availability, verifies user credentials, and ensures the request is legitimate and adheres to business rules.



Update Seat Database

Upon successful validation, the system atomically updates the seat inventory, marking seats as occupied for bookings or available for cancellations.



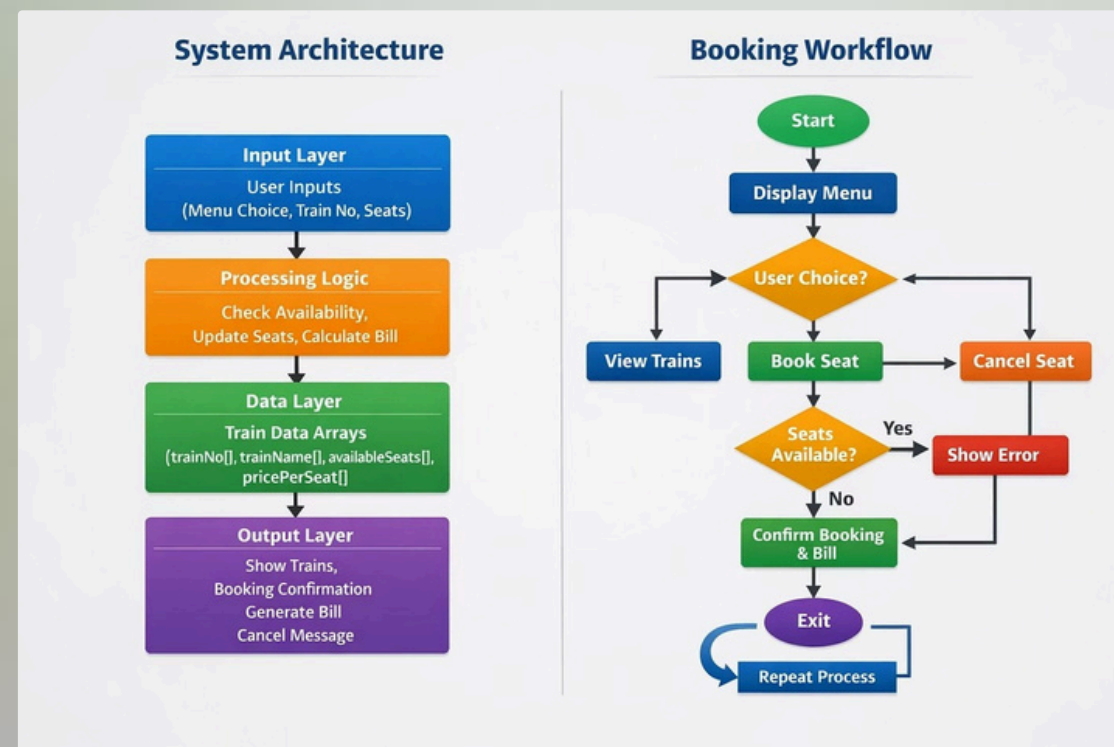
Generate Confirmation

A confirmation message, including booking details and a unique reference number, is presented to the user, ensuring transparency and record-keeping.

This sequential process ensures efficient and accurate management of railway reservations, from initial user input to final confirmation.

System Architecture & Work Flow

Our Railway Reservation System is built upon a modular architecture, designed for clarity, efficiency, and ease of maintenance. Key components work in harmony to deliver a seamless user experience.



Data Storage (Arrays)

Efficiently manages train schedules, seat inventories, and user information using in-memory arrays for quick access and manipulation.

Menu Interface

Provides a user-friendly, console-based menu for navigation, allowing users to select actions like booking, cancellation, and inquiry with ease.

Validation Engine

Ensures data integrity and business rule compliance by meticulously validating user inputs, seat availability, and reservation requests.

Billing Module

Calculates and presents booking costs, generating clear and concise billing information for confirmed reservations, including detailed breakdowns.

What I LEARNED ? : Key Takeaways

The development of the Railway Reservation System has yielded several significant achievements:

Functional Reservation System

Successfully implemented core functionalities for train search, booking, and cancellation, providing a complete user journey.

Real-time Seat Management

Ensures up-to-the-minute seat availability, preventing overbooking and guaranteeing accurate allocations.

Robust Validation

Incorporates rigorous checks for user inputs and seat requests, maintaining data integrity and system reliability.

Transparent Billing

Generates clear, detailed billing for every transaction, fostering user trust and accountability.

Menu-Driven Interface

Offers an intuitive console experience, making navigation and operations straightforward for all users.

This project successfully demonstrates a comprehensive and efficient solution for basic railway reservation needs, providing a solid foundation for future enhancements.



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Thank You

